

James Nguyen

Software Developer | Technical Artist | Animation Pipeline

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EDUCATION

California State University, Los Angeles

Aug. 2021 - May 2026

Bachelor of Science - Computer Science | GPA: 3.4

Relevant Courses: *Software Engineering, Database Systems, Analysis of Algorithms, Game Development, Web Programming, Networking Protocols, Machine Learning, 3D Animation*

Organizations: *Association For Computing Machinery, Gravitas Animation Society*

Gnomon School of VFX, Hollywood CA

Jun. 2026 - Present

Character Rigging Fundamentals (Spencer Diaz)

SOFTWARE ENGINEERING EXPERIENCE

WaxHeart Team | Technical Artist

Jun. 2026 - Present

Python, C#, Unity, Unity Test Framework (UTF), ShaderGraph

- Developing and optimizing custom shaders in Unity enhancing VFX and performance across game prefabs.
- Working on displacement shaders in Unity URP ShaderGraph, coordinating directly with art leads to match approved visual direction and improve environmental realism.

Los Angeles Bureau of Engineering | Rover Engineering Intern

Sept. 2025 - Apr. 2026

Bureau of Engineering - City of Los Angeles

- Coordinated a multi-sensor IMU module build across accelerometers and an I2C multiplexer, tracking component status and documenting integration steps for software and electrical engineering teams and city liaisons.
- Developed firmware to synchronize accelerometer data acquisition with LiDAR scan events for real-time terrain displacement validation on an autonomous rover platform.
- Achieved 98% pixel accuracy developing a UNet-based semantic segmentation model to detect sidewalk defects from rover LiDAR imagery, improving city infrastructure maintenance.

Badminton Court Reservation System | Full-Stack Developer

Sept. 2023 - Jan. 2024

Spring Boot, React, PostgreSQL, Postman, Heroku, Jira

- Led Agile development and sprint planning with JIRA Scrum, coordinating task tracking and status communication across the engineering team reducing project timeline by 40%.
- Built a full-stack web app enabling user account management and court bookings, with ARIES-based recovery ensuring data consistency for transactional operations.
- Wrote unit and integration tests using JUnit and Postman, achieving 85%+ test coverage across core booking and payment workflows.

ADDITIONAL PROJECTS

Maya → Unreal FBX Exporter Tool | Pipeline Project

Jan. 2026

Python, Autodesk Maya API, Unreal Engine 5

- Scripted a tool streamlining the asset integration workflow, removing need for manual file cleanup and ensuring error-free, ready-to-use engine imports.

TECHNICAL SKILLS

Organization & Pipeline Tools: Git, Jira, VS Code, Eclipse, IntelliJ IDEA, Postman, Selenium

Creative & Graphics Tools: Adobe Photoshop, Adobe Suite, Autodesk Maya, Unity, Unreal Engine 5

Programming Languages: Python, Java, C#, C++, Golang, TypeScript, Kotlin, SQL

Frameworks: React.js, Next.js, Node.js, PostgreSQL, Spring Boot, FastAPI, Svelte, PyTorch